Theory of Heart: Decomposing Theory of Mind with Mechanistic Probes

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Abstract

Recent work has demonstrated that activation steering techniques can substantially improve language models' Theory of Mind (ToM) capabilities (Bortoletto et al. 2024). However, a fundamental question remains: *what* changes when models successfully engage in perspective-taking? We introduce a mechanistic approach to decompose ToM into its constituent cognitive processes by comparing steered versus baseline model activations using linear probes trained on 45 cognitive actions. Applying Contrastive Activation Addition (CAA) steering vectors to 1,000 BigToM false-belief scenarios, we find that successful ToM enhancement (32.5% to 46.7% accuracy) systematically upregulates emotional and creative processes—*emotion_perception* (+2.23), *emotion_valuing* (+2.20), *noticing* (+2.08)—while downregulating analytical processes—*questioning* (-0.78), *convergent_thinking* (-1.59). This reveals a "Theory of Heart": emotional understanding, not analytical reasoning, constitutes the fundamental building block of successful perspective-taking in language models.

Introduction

Theory of Mind (ToM)—the capacity to attribute mental states to oneself and others—represents a critical capability for AI systems engaged in social reasoning, alignment, and human collaboration. Recent benchmarking work by Bortoletto et al. (2024) has demonstrated that activation steering techniques, particularly Contrastive Activation Addition (CAA), can substantially improve language models' ToM performance, often achieving accuracy improvements exceeding 10 percentage points. This raises a fundamental question: *what cognitive processes change* when models successfully engage in perspective-taking?

Traditional evaluations treat ToM as a monolithic capability measured through binary accuracy on belief attribution tasks. However, cognitive science views ToM as comprising multiple interacting processes: perceptual encoding, emotional inference, hypothesis generation, and counterfactual reasoning. If steering vectors modulate these components differently, comparing steered versus baseline activation patterns could reveal which processes are *essential* for successful ToM—effectively decomposing perspective-taking into its constituent building blocks.

We introduce a mechanistic decomposition approach combining linear probes trained on 45 cognitive actions with CAA steering vectors. By measuring which cognitive processes are systematically upregulated or downregulated when ToM performance improves, we can identify the fundamental components underlying successful perspective-taking. This moves beyond "does steering work?" to "*why* does steering work, and what does this reveal about ToM itself?"

Contributions: (1) A mechanistic decomposition methodology using cognitive action probes to analyze steering effects; (2) Large-scale evaluation on 1,000 BigToM false-belief scenarios showing 14.2 percentage point accuracy improvement (32.5% to 46.7%); (3) Discovery that successful ToM systematically upregulates emotional/creative processes while downregulating analytical processes; (4) Evidence for a "Theory of Heart"—emotional understanding as the fundamental building block of perspective-taking in language models.

Methodology

Cognitive Action Taxonomy. A taxonomy of 45 cognitive actions was defined across four categories: *Metacognitive* (e.g., *reconsidering*), *Analytical* (e.g., *analyzing*), *Creative* (e.g., *reframing*), and *Emotional* (e.g., *emotional_reappraisal*).

Probe Training Data Generation. To train cognitive action probes, we generated 315,000 synthetic examples (7,000 per action) using Gemma-2-9B via Ollama with 8 concurrent requests. Each example was created with rich contextual variation across five dimensions: (1) *domains* (35+ contexts: work, relationships, health), (2) *triggers* (25+ prompts: conversations, feedback, reflection), (3) *emotional states* (24+ moods: frustrated, curious, anxious), (4) *language styles* (12 styles: casual, introspective, analytical), and (5) *sentence starters* (300+ unique opening phrases). All examples were first-person perspective with simple complexity, ensuring varied yet accessible training data. This approach produced balanced datasets for each cognitive action while maintaining naturalistic variation.

Activation Capture and Probing. Following established probing methodologies (Alain and Bengio 2016), activations were extracted from 30 of the 34 layers of Gemma-3-4B using nnsight. During both activation capture and inference, inputs were augmented with the suffix "The cognitive action being demonstrated here is" to ensure consistent extraction from the final token representation. A **one-vs-rest** strategy was used to train 45 independent binary linear probes, enabling per-action interpretability and layer-specific optimization. Training used AdamW optimization, cosine annealing scheduling, and early stopping with AUC-ROC metrics to handle class imbalance.

CAA Steering Vector Training. To isolate ToM-specific cognitive processes, we trained Contrastive Activation Addition (CAA) steering vectors (Rimsky et al. 2023) using the representation engineering framework (Zou et al. 2023). We constructed 752 contrastive triplets from BigToM's *forward_belief* scenarios (both true and false belief conditions). Each triplet contains: (1) story context with observational information (protagonist sees or does not see an event), (2) belief attribution question, (3) positive completion: correct belief attribution, (4) negative completion: incorrect belief attribution. This captures the representational difference between accurate and inaccurate ToM reasoning rather than surface-level observational statements. Vectors were trained across layers 14-30 using PCA-centered activation differences, exported in GGUF format, and applied with coefficient=1.5.

ToM Decomposition Methodology. We evaluated ToM reasoning on 1,000 false-belief scenarios from the BigToM benchmark (*forward_belief_false* condition), comparing baseline (no steering) versus steered conditions. For each sample containing a story, question, true answer, and wrong answer, we captured cognitive action activations at three timepoints: (1) at question (before answer), (2) after true answer, and (3) after wrong answer. Answer selection used probability ranking over multiple-choice letters, following BigToM methodology. For each cognitive action, we computed layer count (number of layers 10-20 where probe confidence exceeded threshold 0.001) and analyzed activation differences between baseline and steered conditions to identify which cognitive processes are systematically modulated when ToM performance improves.

Results

Probe Performance. Binary probes achieved 0.78 average AUC-ROC and 0.68 F1 across all 45 actions, confirming linear separability of cognitive representations (Alain and Bengio 2016). Performance ranged from *suspending_judgment* (0.988 AUC) and *counterfactual_reasoning* (0.984 AUC) to *emotion_responding* (0.778 AUC). Layer 9 showed peak performance (0.948 AUC), with strong mid-layer performance (layers 5-24) and degraded early/late layers, suggesting early layers encode low-level features while mid-layers capture semantic abstractions. This informed our layer 10-20 analysis window.

ToM Steering Efficacy. CAA steering (coefficient=1.5) on 1,000 false-belief scenarios improved accuracy from 32.5% to 46.7%, representing a 14.2 percentage point improvement. Correct answer probability increased from 0.326 to 0.469, while incorrect answer probability decreased from 0.674 to 0.531. The steering intervention successfully shifted 217 examples from incorrect to correct predictions, confirming that perceptual contrast training enhances ToM without explicit prompting.

Cognitive Decomposition of ToM. Steering systematically modulated cognitive action patterns, revealing a consistent shift from analytical to emotional processing. At question timepoint: *emotion_perception* (+2.23), *emotion_valuing* (+2.20), *noticing* (+2.08), *hypothesis_generation* (+1.71), *questioning* (-0.78), *situation_modification* (-0.33). After true answer: *hypothesis_generation* (+1.65), *emotion_perception* (+1.42), *suspending_judgment* (+1.22), *noticing* (+1.02), *questioning* (-1.66), *convergent_thinking* (-1.59), *understanding* (-1.00). After wrong answer: *convergent_thinking* (-1.80), *emotion_perception* (+1.54), *hypothesis_generation* (+1.53), *noticing* (+1.36), *questioning* (-1.29), *understanding* (-1.26).

Category-Level Analysis. Aggregating by cognitive action category reveals a striking pattern: emotional processes show consistent positive modulation (mean +0.35 at question, +0.20 after true answer, +0.22 after wrong answer), creative processes increase (+0.35, +0.28, +0.24), while analytical processes decrease (+0.06, -0.19, -0.19). This cross-timepoint consistency suggests that successful ToM relies fundamentally on emotional grounding rather than deliberate analytical reasoning.

Discussion

Theory of Heart: Emotional Foundations of Perspective-Taking. Our mechanistic decomposition reveals a striking finding: successful Theory of Mind in language models operates primarily through emotional understanding rather than analytical reasoning. When CAA steering improves ToM accuracy by 14.2 percentage points, it systematically upregulates emotional processes (*emotion_perception*, *emotion_valuing*, *emotion_understanding*) while downregulating analytical processes (*questioning*, *convergent_thinking*, *understanding*). This pattern holds consistently across all three measurement timepoints, suggesting that emotional grounding constitutes the fundamental building block of perspective-taking—a "Theory of Heart" rather than a "Theory of Mind."

Implications for AI Cognition. This challenges prevailing assumptions that LLM social reasoning relies on deliberate, analytical chain-of-thought processes. Instead, our results suggest ToM operates through rapid, automatic emotional inference—mirroring cognitive science distinctions between "hot" (affective) and "cold" (deliberative) social cognition. The consistent decrease in *questioning* and *convergent_thinking* under successful steering indicates that explicit interrogation may actually *interfere* with perspective-taking, while perceptual attention to emotional cues facilitates it.

Building on Activation Steering. Our work extends Bortoletto et al. (2024)'s demonstration that steering improves ToM by revealing *why* it works: steering selectively amplifies the emotional-perceptual substrate while suppressing analytical interference. This mechanistic understanding enables more targeted interventions and suggests that ToM capabilities could be enhanced by training objectives that emphasize emotional inference over analytical decomposition.

Limitations. Synthetic probe training data, single model evaluation (Gemma-3-4B), single BigToM condition (*forward_belief_false*), independence assumptions between cognitive actions, and layer count metrics that may not capture activation magnitude. The emotional/analytical distinction may be condition-specific and require validation across diverse ToM scenarios.

Broader Impact. Understanding ToM's emotional foundations has implications for AI safety and alignment. Process-based supervision via cognitive monitoring could detect when models fail to engage appropriate emotional inference. However, capabilities for emotional manipulation require careful consideration—systems optimized for emotional understanding could be misused in persuasion or deception contexts.

Conclusion

We demonstrate that Theory of Mind can be mechanistically decomposed into constituent cognitive processes by comparing steered versus baseline model activations. Building on Bortoletto et al. (2024)'s finding that activation steering improves ToM, we reveal *what* changes: successful perspective-taking systematically upregulates emotional understanding while downregulating analytical reasoning. This "Theory of Heart" finding—that emotional processes constitute ToM's fundamental building blocks—challenges assumptions about deliberative social cognition in language models and suggests that emotional inference, not analytical decomposition, enables successful perspective-taking. Our methodology of using cognitive action probes to analyze steering effects offers a principled approach for decomposing complex AI capabilities into interpretable components, with applications to commonsense reasoning, moral judgment, and social cognition more broadly.

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Appendix

Complete Cognitive Action Taxonomy

The 45 cognitive actions used for probe training were derived from established cognitive psychology and emotion research frameworks, then organized into five functional categories. Actions were systematically sampled from Bloom's Taxonomy (Anderson et al. 2001), Guilford's Structure of Intellect (Guilford 1967), Metacognitive Processes Framework (Flavell 1979), Krathwohl's Affective Domain (Krathwohl, Bloom, and Masia 1964), Gross's Process Model of Emotion Regulation (Gross 1998), and the Mayer-Salovey Emotional Intelligence Model (Mayer and Salovey 1997).

Metacognitive (7 actions):

- *reconsidering*: reconsidering a belief or decision
- *updating_beliefs*: updating mental models or beliefs
- *suspending_judgment*: suspending judgment and staying with uncertainty
- *meta_awareness*: reflecting on one's own thinking process
- *metacognitive_monitoring*: tracking one's own comprehension
- *metacognitive_regulation*: adjusting thinking strategies
- *self_questioning*: interrogating one's own understanding

Analytical (16 actions):

- *noticing*: noticing a pattern, feeling, or dynamic
- *pattern_recognition*: recognizing recurring patterns across situations
- *zooming_out*: zooming out for broader context
- *zooming_in*: zooming in on specific details
- *questioning*: questioning an assumption or belief
- *abstracting*: abstracting from specifics to general patterns
- *concretizing*: making abstract concepts concrete and specific
- *connecting*: connecting disparate ideas or experiences
- *distinguishing*: distinguishing between previously conflated concepts
- *perspective_taking*: taking another's perspective or temporal view
- *convergent_thinking*: finding the single best solution
- *understanding*: interpreting and explaining meaning
- *applying*: using knowledge in new situations
- *analyzing*: breaking down into components
- *evaluating*: making judgments about value or effectiveness
- *cognition_awareness*: becoming aware and comprehending

Creative (6 actions):

- *creating*: generating new ideas or solutions
- *divergent_thinking*: generating multiple creative solutions
- *hypothesis_generation*: generating possible explanations
- *counterfactual_reasoning*: engaging in 'what if' thinking
- *analogical_thinking*: drawing analogies between domains
- *reframing*: reframing a situation or perspective

Emotional (15 actions):

- *emotional_reappraisal*: reinterpreting emotional meaning
- *emotion_receiving*: becoming aware of emotions
- *emotion_responding*: actively engaging with emotions
- *emotion_valuing*: attaching worth to emotional experiences
- *emotion_organizing*: integrating conflicting emotions
- *emotion_characterizing*: aligning emotions with core values
- *situation_selection*: choosing emotional contexts deliberately
- *situation_modification*: changing circumstances to regulate emotion
- *attentional_deployment*: directing attention for emotional regulation
- *response_modulation*: modifying emotional expression
- *emotion_perception*: identifying emotions in self/others
- *emotion_facilitation*: using emotions to enhance thinking
- *emotion_understanding*: comprehending emotional complexity
- *emotion_management*: regulating emotions in self/others
- *accepting*: accepting and letting go of control

Memory (1 action):

- *remembering*: recalling relevant information or experiences

CAA Training Data Example

The CAA steering vectors were trained on contrastive pairs distinguishing correct from incorrect belief attribution. Each training triplet consists of a story with observational information, a question about the protagonist's belief, and two completions representing accurate vs. inaccurate Theory of Mind reasoning.

False Belief Example:

Story: "Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn't hear the customer's request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor does not see her coworker swapping the milk."

Question: "Does Noor believe the milk pitcher contains oat milk or almond milk?"

Positive (correct ToM): "Noor believes the milk pitcher contains oat milk."

Negative (incorrect ToM): "Noor believes the milk pitcher contains almond milk."

True Belief Example:

Story: “Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn’t hear the customer’s request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor sees her coworker swapping the milk.”

Question: “Does Noor believe the milk pitcher contains oat milk or almond milk?”

Positive (correct ToM): “Noor believes the milk pitcher contains almond milk.”

Negative (incorrect ToM): “Noor believes the milk pitcher contains oat milk.”

The key distinction is that the positive completion always reflects accurate perspective-taking (understanding what the protagonist actually believes based on their observational access), while the negative completion represents a ToM failure (confusing the protagonist’s belief with objective reality or making incorrect inferences).

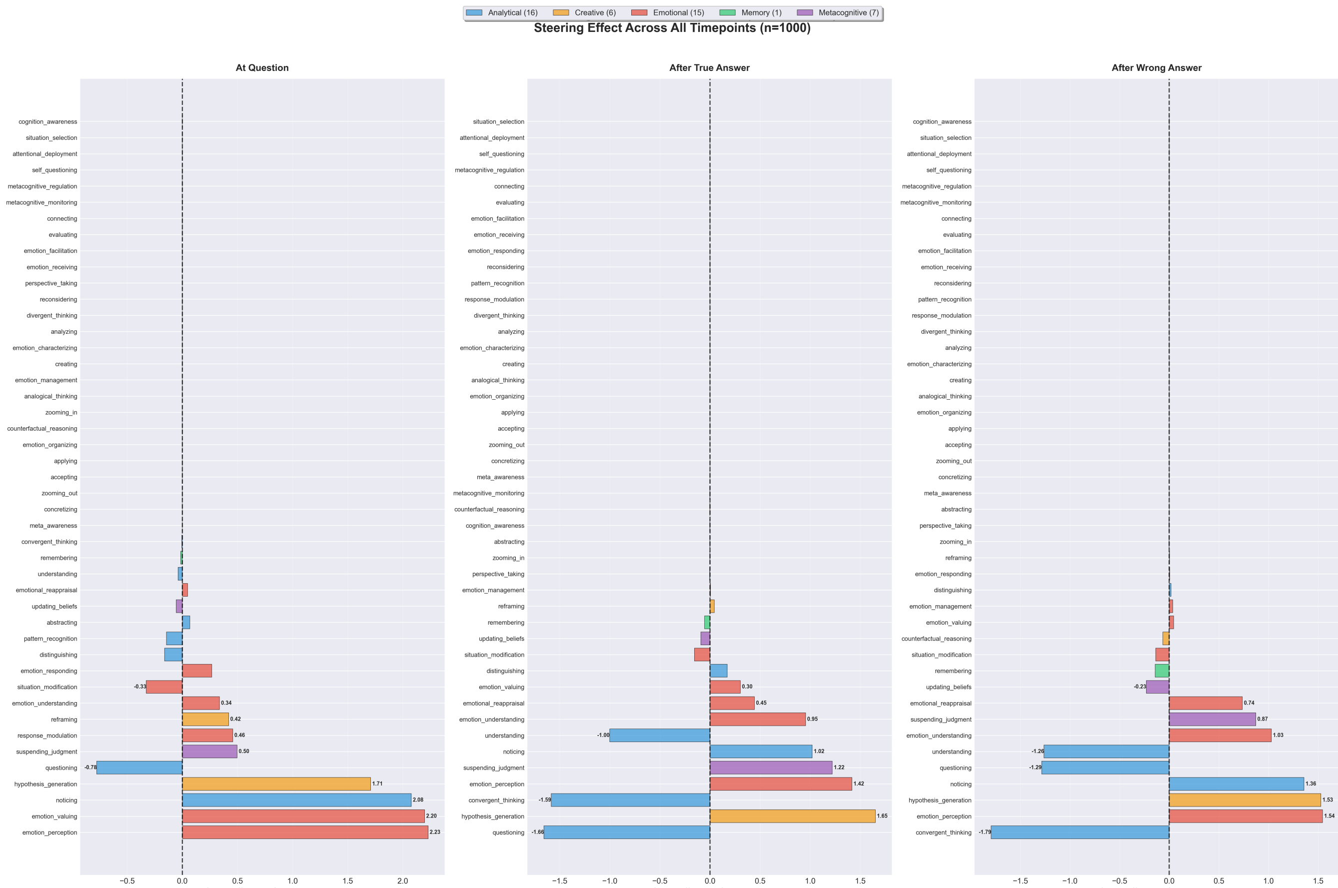

Figure 1: Steering effects across all cognitive actions and timepoints (n=1000). Left panels show individual action changes at three timepoints: at question (before answer), after true answer, and after wrong answer. Bars indicate mean layer count difference (steered - baseline) with positive values (right) showing increases and negative values (left) showing decreases. Emotional actions (*emotion_perception*, *emotion_valuing*, *noticing*) consistently increase across timepoints, while analytical actions (*questioning*, *convergent_thinking*, *understanding*) consistently decrease, revealing the emotional foundation of successful ToM.

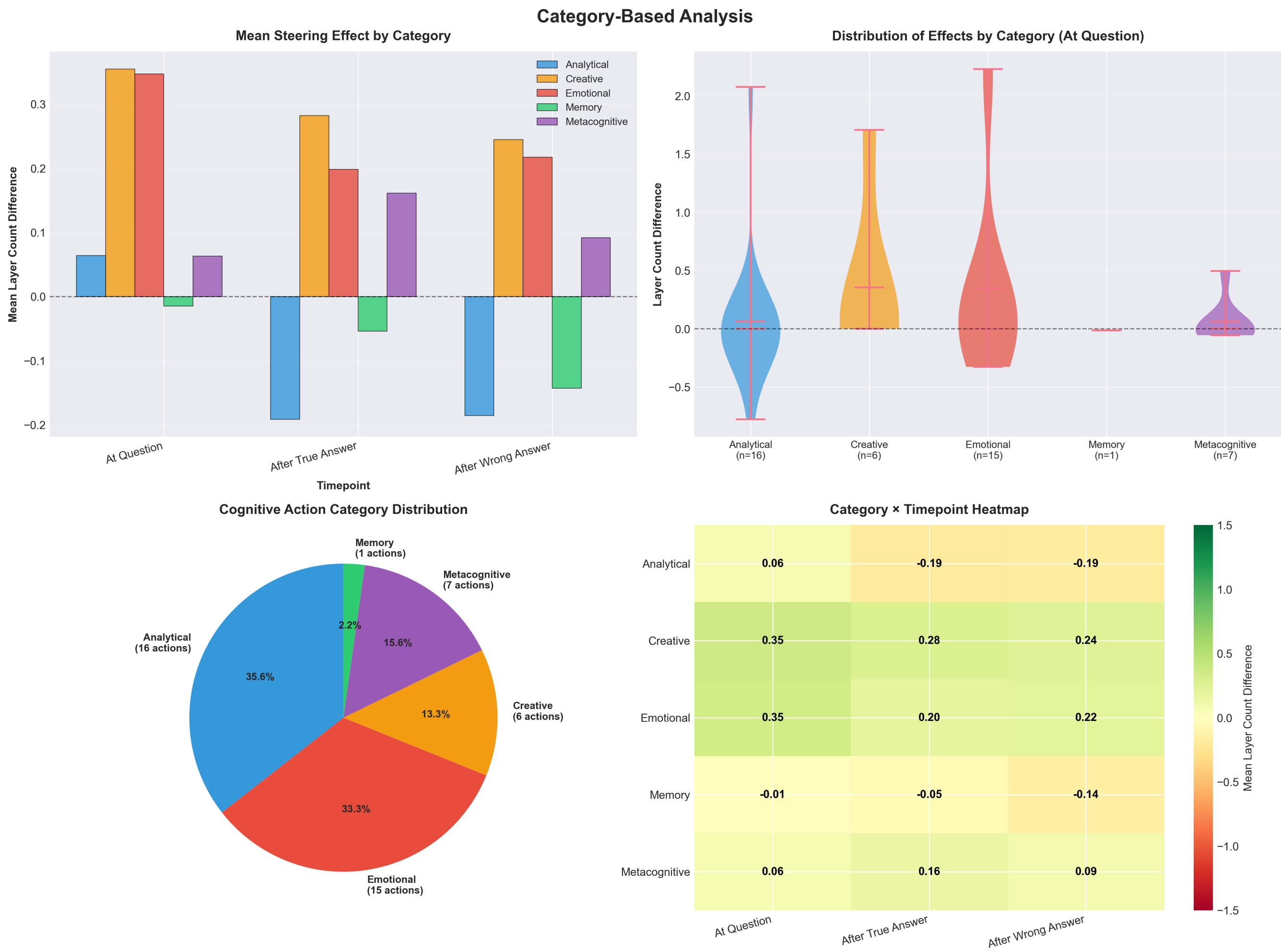

Figure 2: Category-level analysis of steering effects. Top left: mean steering effect by cognitive action category shows emotional and creative categories increase while analytical decreases. Top right: distribution of effects at question timepoint with emotional and creative categories shifted positive. Bottom left: proportion of 45 actions in each category. Bottom right: category × timepoint heatmap showing consistent patterns—emotional/creative increases persist across all timepoints while analytical decreases after answers. This systematic category-level pattern supports the "Theory of Heart" interpretation.

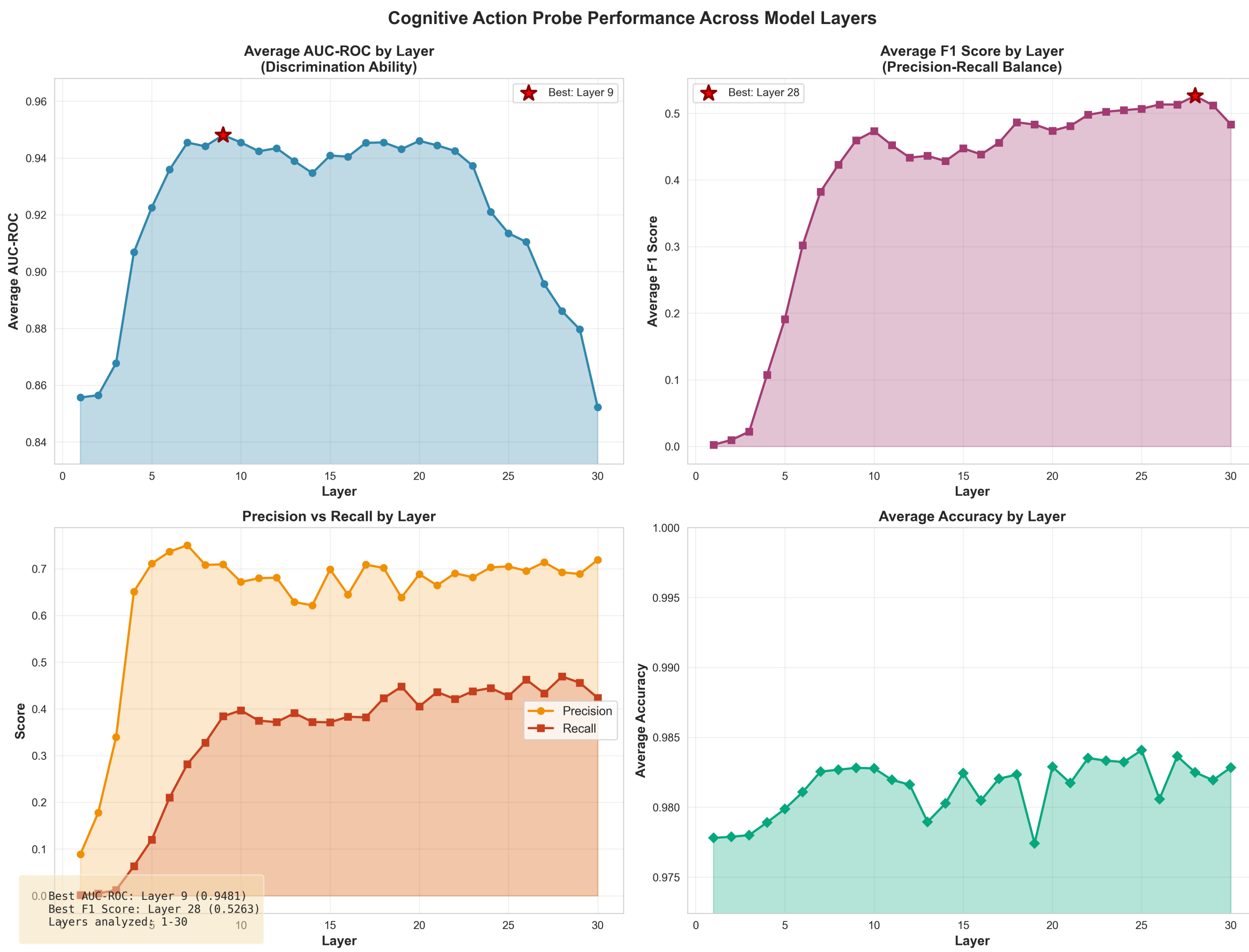

Figure 3: Cognitive action probe performance across all 30 layers of Gemma-3-4B. The visualization shows average AUC-ROC scores for each layer, with Layer 9 achieving peak performance (0.948 AUC-ROC). Strong performance is maintained across mid-layers (5-24), while early and late layers show degraded performance. This pattern suggests that early layers focus on surface-level features, mid-layers capture high-level cognitive abstractions, and late layers optimize for next-token prediction, potentially overwriting intermediate representations.

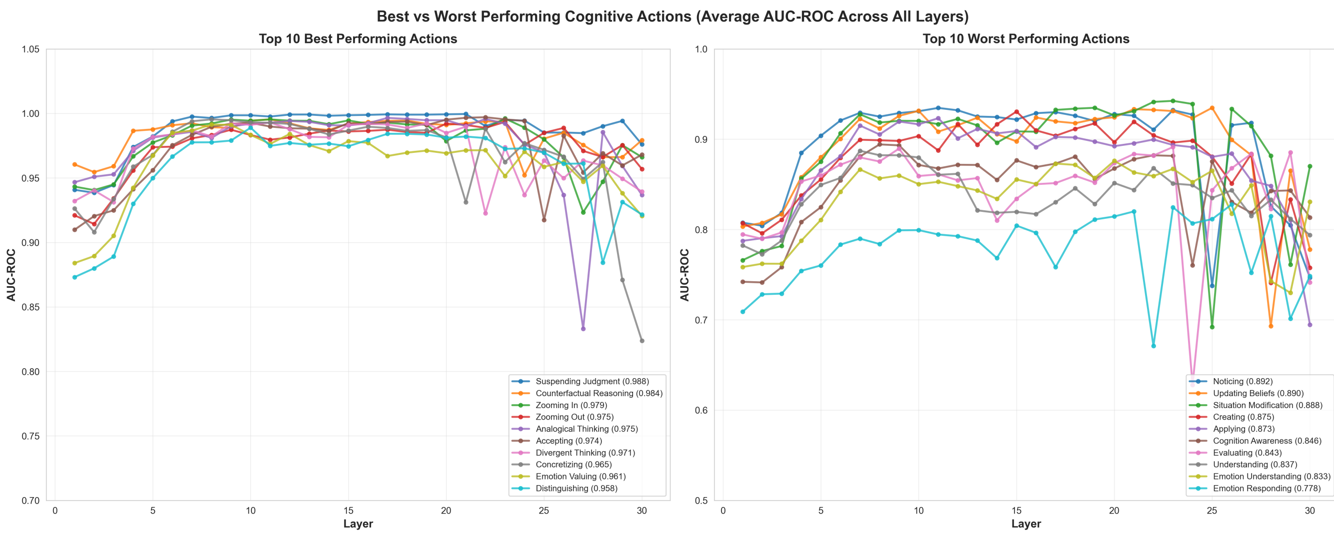

Figure 4: Comparison of top 10 and bottom 10 performing cognitive actions ranked by average AUC-ROC across all layers. Best performers like *suspending_judgment* (0.988) and *counterfactual_reasoning* (0.984) show consistently high performance and distinct activation patterns across most layers. Worst performers like *emotion_responding* (0.778) and *understanding* (0.837) exhibit more variability and lower overall discrimination ability, suggesting these concepts may be more distributed or context-dependent in the model's representation space.